Learning to decode correlated quantum errors: a topological graph neural decoder that surpasses minimum-weight matching under structured circuit-level noise

Molena Huynh[1, *]

[1] *North Carolina State University*

(Dated: July 1, 2026)

Statistics is not a diagnostic bolted onto quantum computing but structural to it: a quantum computer is interrogated only by sampling, and decoding a quantum error-correcting code is, in this light, fundamentally a problem of statistical inference—a syndrome is a noisy observation, and the correct logical class is the one the fault distribution makes most probable, so the decoder that a fault-tolerant machine actually needs is a Bayesian estimator of a posterior it can only learn from samples. We show that the standard surface-code decoder is provably suboptimal under realistic correlated noise, and we close the gap with a learned decoder that never sees the noise model. Minimum-weight perfect matching (MWPM) scores a correction by summing per-edge weights of a graphlike error model, an independence assumption that hardware violates: crosstalk and correlated two-qubit faults flip several detectors jointly, and such hyperedge faults are exactly what a sum of independent edge weights cannot represent. We prove that MWPM is maximum-likelihood if and only if the fault distribution factorizes over graphlike edges, and we construct correlated instances on which the MWPM logical error rate is bounded away from optimal by the probability mass on logical-flipping hyperedges. We then introduce a topological graph neural decoder, a message-passing network on the detector graph with one node per detector and edges joining detectors that share a fault mechanism, so correlated hyperedge faults appear as denser local subgraphs. Trained only on samples, it estimates the Bayes posterior over the logical class and returns a calibrated confidence. On Stim circuit-level rotated surface-code memory experiments decoded head-to-head on identical shots, the learned decoder is statistically indistinguishable from MWPM under independent noise and cuts the logical error rate by 46.1% at distance 3 under correlated noise (paired McNemar $p = 2.4 \times 10^{-164}$), recovering most of the gap to the model-aware BP+OSD ceiling without being given the model. The advantage scales with correlation strength and code distance and is largest in the near-term regime. We report Wilson and paired-bootstrap intervals and release the full pipeline.

I. INTRODUCTION

A quantum computer is, from the outside, a stochastic device: the Born rule returns one sample from a distribution per run, so nearly everything useful about it—estimating observables, certifying gates, characterizing noise—is a statistical task, and error correction is no exception. Fault-tolerant quantum computing rests on a classical inference problem: turning a stream of stabilizer measurements into a correction. This is Bayesian inference in the strict sense—given a syndrome, infer the most probable logical class under the fault distribution—and the quality of that inference, not the raw device physics, sets the achievable logical error rate. Quantum error correction (QEC) encodes logical information across many physical qubits and repeatedly measures stabilizers, whose outcomes form a *syndrome* [1–3, 27–29]. A threshold theorem guarantees that, below a critical physical error rate, arbitrarily long computations can be protected [31]. Topological codes—toric and surface codes [4–6, 30]—are the leading route to scalable fault tolerance, with thresholds set by an underlying statistical-mechanical phase transition [32]; related topological color codes offer transversal gate sets [33]. Recent hardware has driven the logical error rate below the physical one by scaling a surface code [7, 43], following repeated-correction demonstrations at distance three [42]. The component that converts a syndrome into a correction is the *decoder*, where classical algorithms and machine learning meet device physics, and its quality directly sets the achievable logical error rate.

The dominant surface-code decoder is minimum-weight perfect matching (MWPM) [5, 6, 8, 39], built on Edmonds' blossom algorithm [9] and, in practice, on the detector error model (DEM) that a tool such as Stim [10] extracts from a noisy circuit. Alternative graphlike decoders such as union-find trade a small accuracy loss for near-linear runtime [38], and exact maximum-likelihood decoders are known but scale unfavourably with code size [34, 36]. MWPM is fast, near-optimal under independent noise, and has excellent thresholds. Its optimality, however, hinges on one structural assumption: it represents the error model as a *graph*, with one weighted edge per fault that flips at most two detectors, and it scores a correction by *summing* those weights. This coincides with maximum-likelihood decoding when faults are independent and graphlike—and *only* then. In statistical terms, matching commits to a factorized model of the fault distribution; where the real distribution has correlations, that hand-specified model is misspecified, and the decision it induces is no longer the Bayes-optimal one.

Hardware noise is neither independent nor graphlike. Crosstalk, leakage, and correlated two-qubit gate

* molena.huynh@jmp.com

faults produce errors that flip three or four detectors jointly [7, 19]—each such correlated fault a hyperedge in the DEM. Reweighting the matching graph to absorb likely correlations [37] or estimating those weights adaptively from data [40, 41], and tailoring the code to biased noise [35], mitigate but do not remove this limitation. Matching cannot represent a hyperedge: at best it decomposes the hyperedge into independent graphlike edges and sums their weights as if unrelated, discarding the joint structure. Belief-propagation decoders with ordered-statistics post-processing (BP+OSD) [11, 12] use the full hyperedge model and are the strong baseline for correlated and quantum-LDPC codes, but they require the correct model as input and are heavier at inference. A large literature applies neural networks and reinforcement learning to decoding [13–18, 44–46], including graph- and equivariance-aware architectures matched to the detector structure [47, 48], and learned decoders now exceed matching on hardware syndromes [19]. What has been missing is a controlled, reproducible account of why and by how much a learned decoder helps precisely where the matching independence assumption breaks, evaluated head-to-head against both MWPM and a correctly specified BP+OSD on identical shots.

We treat decoding as inference on the code detector graph and make the matching-learning gap both provable and exploitable. Our contributions are three. First, in Sec. III we prove that MWPM is maximum-likelihood if and only if the fault distribution factorizes over graphlike edges, and we exhibit correlated instances on which the MWPM logical error rate exceeds that of the optimal decoder by at least the probability mass on a logical-flipping hyperedge it cannot represent—the formal origin of the empirical gap. Second, in Sec. V we introduce a topological graph neural decoder, a message-passing network on the detector graph in which detectors are nodes and fault co-occurrence supplies edges, so correlated hyperedges become denser subgraphs; trained only on samples, it outputs the Bayes posterior over the logical class and yields a calibrated confidence for selective decoding. Third, in Sec. VI we evaluate on Stim circuit-level rotated surface-code memory experiments across distances $3, 5, 7$ and two noise regimes, with all decoders scoring identical shots under paired McNemar and bootstrap statistics: the learned decoder matches MWPM under independent noise and beats it under correlated noise, closing most of the gap to BP+OSD without knowing the model. Every reported number is computed by the released pipeline and traced to a single `summary.json`, with Wilson and paired-bootstrap intervals.

This manuscript belongs to a single-author program on spectral truncation of operators for query-efficient and resource-accounted quantum computation, and it carries forward that program's CPU-reproducible, audit-gated style. The earlier entries develop graph-conditioned trust regions and their calibrated and extended variants for quantum approximate optimization [55, 56], the measurement and query-budget accounting of warm-started low-depth optimization [57, 58], topology-conditioned parameter transfer [59], operator-spectral truncated priors and spectral-truncation graph kernels for schedule transfer [60, 61], and the commutator-graph and Lie-algebraic spectral-truncation treatments of product-formula error [62, 63]. The present work carries that same operator-graph viewpoint into quantum error correction, where the detector graph plays the role of the truncated operator structure and correlated fault mechanisms appear as its denser local subgraphs. The Hamiltonian-simulation work of Nguyen and Jing supplies complementary fixed-depth and representation-theoretic context for the circuits whose errors QEC must suppress [64, 65].

II. BACKGROUND

a. Rotated surface code and the memory experiment. We study the distance-d rotated surface code, the standard planar layout with d^2 data qubits and d^2-1 weight-two and weight-four stabilizers. We run the canonical Z-basis memory experiment: prepare the logical $|0\rangle$, idle for $T = d$ rounds of noisy stabilizer extraction, and read out the logical Z operator. Bit-flip (X) errors anticommute with the Z-stabilizers and are the events to be decoded; a run *fails* if the recovered logical value differs from the prepared one. All circuits, including the syndrome-extraction gate schedule and the measurement-and-reset of ancillas, are generated by Stim [10], so the geometry and fault propagation are exact.

b. Detectors, observables, and the detector error model. A *detector* is a parity of measurement outcomes that is deterministic in the absence of noise—here, the difference between a stabilizer's value in consecutive rounds—so a lit detector marks a space-time location where the syndrome changed. The *logical observable* is the parity that encodes the decoded logical bit. Stim compiles a circuit into a *detector error model* (DEM): a list of independent fault mechanisms, each firing with a probability and flipping a fixed set of detectors and (possibly) the observable. A mechanism that flips at most two detectors is *graphlike*—an edge in a decoding graph; one that flips three or more is a *hyperedge*. Stim can emit the DEM in two forms: *decomposed*, where hyperedges are split into graphlike pieces (the view matching consumes), and *full*, where hyperedges are kept intact (the view BP+OSD and our decoder consume).

c. Decoders. Given the per-shot detector pattern, MWPM (PyMatching [8], on the decomposed DEM) returns the minimum-weight set of edges reproducing the syndrome; BP+OSD (`ldpc` [11, 12], on the full DEM) runs belief propagation followed by ordered-statistics post-processing and can exploit hyperedges; our learned decoder (Sec. V) reads the detector pattern off the detector graph and predicts the logical class directly. All three predict the same logical observable on identical shots, so their logical error rates are directly comparable.

III. THEORY: WHEN MATCHING IS OPTIMAL, AND WHEN IT IS NOT

a. Setup. A detector error model (DEM) is a finite set of independent fault mechanisms $\{f_1, \dots, f_m\}$. Mechanism f_j fires independently with probability $p_j \in (0, \frac{1}{2})$ and, when it fires, flips a fixed set D_j of detectors and a fixed set L_j of logical observables. A fault configuration is a vector $x \in \mathbb{F}_2^m$ with $x_j = 1$ iff f_j fired. Collecting the detector incidences into $H \in \mathbb{F}_2^{n_d \times m}$ (column j is the indicator of D_j) and the observable incidences into $L \in \mathbb{F}_2^{k \times m}$, the observed *syndrome* is $s = Hx$ and the *logical flip* is $\ell = Lx \pmod 2$. A decoder sees s and outputs $\hat{\ell}$; it *fails* on x iff $\hat{\ell} \neq Lx$. A mechanism is *graphlike* if $|D_j| \leq 2$. Minimum-weight matching assigns each graphlike mechanism the weight $w_j = \log \frac{1-p_j}{p_j} > 0$ and, on syndrome s, returns $\hat{x}_{\mathrm{MWPM}} = \arg\min_{x:\, Hx=s} \sum_j w_j x_j$, predicting $\hat{\ell} = L\hat{x}_{\mathrm{MWPM}}$; a correlated hyperedge ($|D_j| \geq 3$, or two mechanisms that fire jointly) is not a single weighted edge and is in practice decomposed into graphlike edges whose weights are summed independently.

Lemma 1 (Additive log-likelihood $\Leftrightarrow$ independence)**.** *Let μ be a fully supported distribution on $\mathbb{F}_2^m$. There exist constants C and $w_1, \dots, w_m$ with $-\log \mu(x) = C + \sum_j w_j x_j$ for all $x \in \mathbb{F}_2^m$ if and only if the coordinates $x_1, \dots, x_m$ are mutually independent under μ.*

Proof. ($\Leftarrow$) If the coordinates are independent Bernoulli(p_j) then $\mu(x) = \prod_j p_j^{x_j}(1-p_j)^{1-x_j} = \big(\prod_j (1-p_j)\big) \prod_j \big(\frac{p_j}{1-p_j}\big)^{x_j}$, so $-\log \mu(x) = -\sum_j \log(1-p_j) + \sum_j x_j \log \frac{1-p_j}{p_j}$, affine in x with $w_j = \log \frac{1-p_j}{p_j}$. ($\Rightarrow$) If $-\log \mu(x) = C + \sum_j w_j x_j$ then $\mu(x) = e^{-C} \prod_j e^{-w_j x_j}$. With $q_j = \frac{e^{-w_j}}{1+e^{-w_j}}$, $\prod_j q_j^{x_j}(1-q_j)^{1-x_j} \propto e^{-w \cdot x} \propto \mu(x)$; both are normalized, so $\mu(x) = \prod_j q_j^{x_j}(1-q_j)^{1-x_j}$ and the coordinates are independent. □

Proposition 2 (Matching is maximum-likelihood iff the model factorizes)**.** *Let all mechanisms be graphlike. The additive matching objective coincides with the negative log-likelihood—i.e.* $\arg\min_{x:Hx=s} \sum_j w_j x_j = \arg\max_{x:Hx=s} \mu(x)$ *for every syndrome s and some fixed weights w—if and only if the fault distribution μ factorizes over mechanisms (independence). Under independence the minimum-weight configuration is therefore a maximum-likelihood fault configuration; under any correlated μ no fixed weights make matching maximum-likelihood.*

Proof. By Lemma 1, fixed weights w with $\sum_j w_j x_j = -\log \mu(x) - C$ for all x exist iff μ factorizes. When they exist, minimizing $\sum_j w_j x_j$ on each fiber $\{x : Hx = s\}$ maximizes $\mu(x)$, so matching is maximum-likelihood for every s. Conversely, if matching with fixed w is maximum-likelihood for every s, comparing the empty configuration to singletons $x = e_j$ forces $w_j = \log \frac{1-q_j}{q_j}$ for the marginal $q_j = \mu(x_j{=}1)$, and consistency on the two-fault fibers requires $\mu(e_i{+}e_j)\mu(0) = \mu(e_i)\mu(e_j)$ for all i, j; iterating over larger fibers yields full independence. Hence matching is maximum-likelihood for all syndromes only if μ factorizes. □

Proposition 3 (A provable separation under correlation)**.** *There is a DEM with a single correlated mechanism on which matching's logical error rate exceeds the optimal (Bayes) decoder's by a strictly positive amount. Take two detectors d_1, d_2 and three mechanisms: f_A flips $\{d_1\}$ with probability p_A (no logical), f_B flips $\{d_2\}$ with probability p_B (no logical), and a correlated f_C flips $\{d_1, d_2\}$ and the logical with probability p_C. For the syndrome $s^\star = (1,1)$,*

$$\mathrm{LER}_{\mathrm{MWPM}} - \mathrm{LER}_{\mathrm{opt}} \;\geq\; p_C(1-p_A)(1-p_B) - p_A p_B (1-p_C),$$

which is positive whenever $p_C > \dfrac{p_A p_B}{(1-p_A)(1-p_B) + p_A p_B}$.

Proof. Matching is given the decomposed model: f_C, flipping $\{d_1, d_2\}$, is indistinguishable at the edge level from the pair $\{f_A, f_B\}$, so the graph has only the two boundary edges. On $s^\star = (1,1)$ the unique graphlike explanation is $\{f_A, f_B\}$, of logical value 0; matching therefore predicts $\hat{\ell} = 0$ deterministically. The two ways to light both detectors with opposite logical parity have masses $\Pr[s^\star, \ell{=}1] = p_C(1-p_A)(1-p_B)$ (only f_C fired) and $\Pr[s^\star, \ell{=}0] = p_A p_B (1-p_C)$ (only f_A, f_B fired). Matching errs on all of $\Pr[s^\star, \ell{=}1]$; the Bayes decoder predicts the larger mass and, when $\Pr[s^\star, \ell{=}1] > \Pr[s^\star, \ell{=}0]$, errs only on $\Pr[s^\star, \ell{=}0]$. Restricting to $s^\star$ gives the bound; positivity is the displayed condition. □

Remark 1 (What the learned graph decoder can represent)**.** *The Bayes-optimal predictor $\hat{\ell}(s) = \arg\max_\ell \Pr[Lx = \ell \mid Hx = s]$ is a function of the entire detector pattern under the true law μ, and Proposition 3 shows it strictly dominates matching whenever μ is correlated. A message-passing network on the detector graph is a permutation-equivariant function of that pattern and, with sufficient width and data, approximates any such posterior; in particular it can represent the correlated explanation f_C that matching's additive edge objective cannot. It learns this posterior directly from samples of μ—exactly the information matching is denied when handed only marginal edge weights. This is the formal reason the learned decoder matches matching where μ is graphlike (Proposition 2) and beats it where μ is correlated (Proposition 3), as the experiments confirm.*

IV. DECODING AS BAYESIAN INFERENCE: OPTIMALITY, EXPRESSIVITY, GENERALIZATION

The previous section pinned the matching–learning gap to a representational fact about a *fixed* objective. We now place the whole decoding problem on a statistical footing and answer three questions the empirical study raises. Is there a well-defined optimal decoder that the learned network is trying to imitate, and does the network's output—a posterior over the logical class— target it (Sec. IV A)? Can a message-passing network on the detector graph, of the lightweight kind we deploy, actually represent that optimal decoder (Sec. IV B)? And does training on finitely many shots recover it, with a controlled gap between the error we measure on test data and the error the decoder will incur in deployment (Secs. IV C–IV D)? The four subsections give, respectively, a Bayes-optimality (MAP/coset) characterization, a universal-approximation theorem on the detector graph, a Rademacher/VC generalization bound with an explicit sample complexity, and a risk-consistency theorem. Together they formalize the claim that a learned decoder is a probabilistic surrogate for an intractable maximum-likelihood decoder that, with enough data, provably matches or surpasses matching under structured noise. Throughout we keep the notation of Sec. III: $x \in \mathbb{F}_2^m$ is the fault configuration drawn from the true law μ, $s = Hx$ is the syndrome, $\ell = Lx$ the logical flip; here $k = 1$ (a single logical observable, as in the memory experiment), so $\ell \in \mathbb{F}_2 = \{0, 1\}$.

A. The coset MAP decoder is the Bayes-optimal target

Decoding is a classification problem: from the observed syndrome s predict the logical class $\ell \in \{0, 1\}$. Because two fault configurations with the same syndrome and the same logical value are indistinguishable *and equally good* for the memory experiment—only ℓ, not x, determines success—the correct object to condition on is not a single fault configuration but the coset of configurations carrying a given logical value.

Definition 1 (Syndrome fibers, logical cosets, and posterior)**.** *Fix a DEM (H, L, μ). For a syndrome s in the image of H, the* fiber *is $\Phi(s) = \{x \in \mathbb{F}_2^m : Hx = s\}$, and for $\ell \in \{0, 1\}$ the* logical coset *is $\Phi_\ell(s) = \{x \in \Phi(s) : Lx = \ell\}$. The* coset (class) posterior *is*

$$\eta_\ell(s) \;:=\; \Pr_{x\sim\mu}\big[Lx = \ell \,\big|\, Hx = s\big] \;=\; \frac{\sum_{x\in\Phi_\ell(s)} \mu(x)}{\sum_{x\in\Phi(s)} \mu(x)}. \tag{1}$$

Write $\eta(s) := \eta_1(s)$, so $\eta_0(s) = 1 - \eta(s)$. A decoder is any measurable map $g : \mathcal{S} \to \{0, 1\}$ from syndromes $\mathcal{S} := \mathrm{im}(H)$ to logical classes; its logical risk *(expected logical error rate) is*

$$\mathcal{R}(g) \;:=\; \Pr_{x\sim\mu}\big[\, g(Hx) \neq Lx \,\big] \;=\; \mathbb{E}_s\big[\, \eta_{1-g(s)}(s) \,\big], \tag{2}$$

where $s = Hx$ has the pushforward law $\nu := H_\#\mu$ on $\mathcal{S}$. The coset MAP *(Bayes) decoder is $g^\star(s) := \arg\max_{\ell\in\{0,1\}} \eta_\ell(s) = \mathbf{1}\big[\eta(s) > \frac{1}{2}\big]$.*

Proposition 4 (Coset MAP is Bayes-optimal; MWPM is fiber-MAP, not coset-MAP)**.** *For every DEM (H, L, μ) and every decoder g,*

$$\mathcal{R}(g) - \mathcal{R}(g^\star) \;=\; \mathbb{E}_{s\sim\nu}\Big[\, \big|2\eta(s) - 1\big|\, \mathbf{1}[g(s) \neq g^\star(s)] \,\Big] \;\geq\; 0, \tag{3}$$

so $g^\star$ attains the minimum logical risk $\mathcal{R}^\star := \mathcal{R}(g^\star) = \mathbb{E}_s\big[\min\{\eta(s), 1 - \eta(s)\}\big]$, and it is the unique minimizer up to ν-null sets where $\eta(s) = \frac{1}{2}$. Moreover the maximum-likelihood configuration decoder $g_{\mathrm{ML}}(s) := L\,\hat{x}(s)$ with $\hat{x}(s) \in \arg\max_{x\in\Phi(s)} \mu(x)$ can differ from $g^\star$: it maximizes mass over a single *configuration rather than over a coset, and MWPM is the special case in which $-\log\mu$ is replaced by the additive surrogate $\sum_j w_j x_j$ of Proposition 2.*

Proof. Condition on s. For a fixed decision $g(s) = b \in \{0, 1\}$ the conditional error probability is $\Pr[Lx \neq b \mid Hx = s] = \eta_{1-b}(s)$. Summing the identity $\eta_{1-b}(s) - \eta_{1-g^\star(s)}(s) = |2\eta(s) - 1|\,\mathbf{1}[b \neq g^\star(s)]$, which holds because $g^\star(s)$ selects the larger of $\eta_0(s), \eta_1(s)$ and the two conditional errors differ by exactly $|\eta_1(s) - \eta_0(s)| = |2\eta(s) - 1|$ when the decisions differ and by 0 when they agree, and taking $\mathbb{E}_{s\sim\nu}$ gives (3). Each summand is nonnegative, so $\mathcal{R}(g) \geq \mathcal{R}(g^\star)$, with equality iff $g = g^\star$ on $\{s : \eta(s) \neq \frac{1}{2}\}$; on $\{\eta(s) = \frac{1}{2}\}$ the choice is immaterial. The conditional optimal error is $\min\{\eta(s), 1 - \eta(s)\}$, giving $\mathcal{R}^\star$. For the last claim, g_{ML} reads off the logical value of the single most-likely configuration; when a fiber contains a low-mass configuration in one coset but many moderate-mass configurations in the other (*degeneracy*), the coset with the largest total mass (1) need not contain the modal configuration, so $g_{\mathrm{ML}}(s) \neq g^\star(s)$ is possible. MWPM further replaces μ by the factorized surrogate of Lemma 1, so it is g_{ML} for that surrogate, and Proposition 3 is one s on which even the surrogate's fiber-MAP disagrees with $g^\star$. □

Equation (3) is the master identity behind every later claim: the excess risk of *any* decoder over the Bayes-optimal one is the ν-average of the margin $|2\eta(s) - 1|$ over the syndromes it misclassifies. It converts "beat MWPM" into "agree with $g^\star$ on more margin-weighted mass than MWPM does," and it is exactly what a network estimating η then thresholding at $\frac{1}{2}$ is built to do. Summing Proposition 3 over all syndromes makes the population gap explicit.

Corollary 5 (Population logical-error gap of MWPM under correlation)**.** *Let $B := \{s \in \mathcal{S} : g_{\mathrm{MWPM}}(s) \neq g^\star(s)\}$*

be the set of syndromes on which matching disagrees with the coset MAP decoder. Then

$$\mathcal{R}(g_{\mathrm{MWPM}}) - \mathcal{R}^\star \;=\; \sum_{s\in B} \nu(s)\,\big|2\eta(s)-1\big| \;\geq\; 0, \quad (4)$$

and under any μ that factorizes over graphlike edges B is ν-null so the gap vanishes (recovering Proposition 2), whereas for the correlated DEM of Proposition 3 the single syndrome $s^\star = (1,1)$ lies in B and contributes $\nu(s^\star)\,|2\eta(s^\star)-1| > 0$, a strictly positive lower bound on the gap.

Proof. Apply (3) to $g = g_{\mathrm{MWPM}}$; the indicator restricts the sum to B, giving (4). Under factorization Proposition 2 makes matching maximum-likelihood on every fiber and, as its surrogate then equals μ and the memory experiment has $k = 1$, its prediction agrees with $g^\star$ off an $\eta = \frac{1}{2}$ set, so $\nu(B) = 0$. For the correlated DEM, Proposition 3 shows $g_{\mathrm{MWPM}}(s^\star) = 0 \neq 1 = g^\star(s^\star)$ once $p_C > \frac{p_A p_B}{(1-p_A)(1-p_B)+p_A p_B}$, and there $\nu(s^\star) = \Pr[Hx = s^\star] > 0$ and $\eta(s^\star) \neq \frac{1}{2}$, so the displayed term is strictly positive. □

B. A graph network on the detector graph can represent the Bayes decoder

The learned decoder outputs $\hat{q}(s) = \sigma(\mathrm{logit})$ as an estimate of $\eta(s)$. For this to be more than an aspiration the network class must *contain* a function close to η. Because the syndrome space $\mathcal{S} \subseteq \mathbb{F}_2^{n_d}$ is finite, $\eta : \mathcal{S} \to [0,1]$ is an arbitrary bounded function on a finite set, and the only real question is whether the specific message-passing architecture of Sec. V—symmetric-normalized propagation on the fixed detector graph G, ReLU updates, and a global mean/max readout—has the capacity to realize it. Theorem 6 answers yes, and locates where the graph structure helps: distinct detectors carry distinct static coordinates, so the network can separate any two syndromes and then interpolate η exactly.

Theorem 6 (Universal approximation of the coset posterior on the detector graph)**.** *Let G be the detector graph of Sec. V on n_d nodes with the per-node static coordinate features $c_i \in \mathbb{R}^3$ pairwise distinct, and let $\eta : \mathcal{S} \to [0,1]$ be the coset posterior* (1) *of any DEM whose syndromes are read on these nodes. For every $\varepsilon > 0$ there exist a width W, a depth L, and message-passing weights $\{W^{(\ell)}_{\mathrm{self}}, W^{(\ell)}_{\mathrm{neigh}}\}$ with the readout head of Sec. V such that the induced map $\hat{q}_\theta : \mathcal{S} \to (0,1)$ satisfies*

$$\max_{s\in\mathcal{S}} \big|\hat{q}_\theta(s) - \eta(s)\big| \;<\; \varepsilon. \quad (5)$$

Consequently the thresholded decoder $\mathbf{1}\big[\hat{q}_\theta(s) > \frac{1}{2}\big]$ equals the Bayes decoder $g^\star$ on $\{s : |2\eta(s)-1| \geq 2\varepsilon\}$, and its logical risk satisfies $\mathcal{R}(\mathbf{1}\big[\hat{q}_\theta > \frac{1}{2}\big]) \leq \mathcal{R}^\star + \nu(\{s : |2\eta(s)-1| < 2\varepsilon\})$.

The proof (Appendix A) builds the approximant in two stages: the message-passing layers, aided by the distinct coordinates, compute a set of node embeddings from which the mean/max readout is *injective* on the finite syndrome set; a two-layer perceptron then interpolates η on the finitely many resulting readout vectors to accuracy ε by standard ReLU universal approximation. The role of the graph is not approximation power *per se*—a plain MLP on the raw bit-vector is already universal on a finite set—but sample efficiency and locality: correlated hyperedges are dense local subgraphs (Sec. V), so the local message passing that resolves them is exactly where the posterior departs from matching's additive surrogate (Remark 1).

C. PAC generalization: test error controls deployment error

Expressivity is necessary, not sufficient: we fit $\hat{q}$ on a finite sample and must bound the true logical risk by the empirical one. Let $\mathcal{H} = \{h_\theta : \mathcal{S} \to [0,1]\}$ be the class of posterior estimates realized by the architecture of Sec. V over a bounded parameter set, decoded as $g_\theta(s) = \mathbf{1}\big[h_\theta(s) > \frac{1}{2}\big]$. Training draws $S = \{(s_i, \ell_i)\}_{i=1}^n$ i.i.d. from the joint law (Hx, Lx), $x \sim \mu$, and minimizes a surrogate; we bound the 0–1 logical risk. Write the empirical risk $\widehat{\mathcal{R}}_n(g) = \frac{1}{n}\sum_i \mathbf{1}[g(s_i) \neq \ell_i]$ and let $\mathfrak{R}_n(\mathcal{D})$ be the empirical Rademacher complexity of the induced decision class $\mathcal{D} = \{s \mapsto \mathbf{1}\big[h_\theta(s) > \frac{1}{2}\big] : h_\theta \in \mathcal{H}\}$.

Theorem 7 (Rademacher generalization bound for the learned decoder)**.** *For any $\delta \in (0,1)$, with probability at least $1-\delta$ over the draw of S, every g_θ with $h_\theta \in \mathcal{H}$ satisfies*

$$\mathcal{R}(g_\theta) \;\leq\; \widehat{\mathcal{R}}_n(g_\theta) \;+\; 2\,\mathfrak{R}_n(\mathcal{D}) \;+\; 3\sqrt{\frac{\log(2/\delta)}{2n}}. \quad (6)$$

If $\mathcal{D}$ has VC dimension $\mathrm{VC}(\mathcal{D}) =: V < \infty$, then $\mathfrak{R}_n(\mathcal{D}) \leq \sqrt{2V\log(en/V)/n}$, so with probability $\geq 1-\delta$

$$\mathcal{R}(g_\theta) \;\leq\; \widehat{\mathcal{R}}_n(g_\theta) \;+\; 2\sqrt{\frac{2V\log(en/V)}{n}} \;+\; 3\sqrt{\frac{\log(2/\delta)}{2n}}. \quad (7)$$

In particular, to guarantee $\mathcal{R}(g_\theta) \leq \widehat{\mathcal{R}}_n(g_\theta) + \varepsilon$ with confidence $1-\delta$ it suffices that

$$n \;=\; O\!\left(\frac{V\log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}\right). \quad (8)$$

The proof (Appendix B) is the standard symmetrization and bounded- differences (McDiarmid) argument for a $[0,1]$-bounded loss, specialized to the finite syndrome domain, together with the Vapnik–Chervonenkis/Sauer bound relating $\mathfrak{R}_n$ to V; the VC dimension of $\mathcal{D}$ is finite because $\mathcal{H}$ is a fixed-architecture ReLU network with a bounded number of parameters, whose sublevel sets $\{h_\theta > \frac{1}{2}\}$ form a class of finite VC dimension. On

the finite syndrome domain $V \le |\mathcal{S}|$ always, giving a distribution-free fallback.

Corollary 8 (When the learned decoder provably beats MWPM)**.** *Fix a correlated DEM with* $\Delta := \mathcal{R}(g_{\mathrm{MWPM}}) - \mathcal{R}^\star > 0$ *(Corollary 5). Suppose the trained decoder achieves empirical risk* $\widehat{\mathcal{R}}_n(g_\theta) \le \mathcal{R}^\star + \frac{\Delta}{4}$ *(attainable once* $\mathcal{H}$ *approximates* η *well by Theorem 6). If*

$$2\,\mathfrak{R}_n(\mathcal{D}) + 3\sqrt{\tfrac{\log(2/\delta)}{2n}} \;\le\; \tfrac{\Delta}{4}, \tag{9}$$

which by (8) *holds once* $n = O\big((V + \log\frac{1}{\delta})/\Delta^2\big)$*, then with probability at least* $1-\delta$

$$\mathcal{R}(g_\theta) \;\le\; \mathcal{R}^\star + \tfrac{\Delta}{2} \;<\; \mathcal{R}(g_{\mathrm{MWPM}}),$$

i.e. the learned decoder strictly outperforms MWPM by at least $\Delta/2$ *in logical error rate.*

Proof. Combine the empirical-risk hypothesis with (6): with probability $\ge 1-\delta$, $\mathcal{R}(g_\theta) \le \widehat{\mathcal{R}}_n(g_\theta) + 2\mathfrak{R}_n(\mathcal{D}) + 3\sqrt{\log(2/\delta)/(2n)} \le (\mathcal{R}^\star + \Delta/4) + \Delta/4 = \mathcal{R}^\star + \Delta/2$. Since $\mathcal{R}(g_{\mathrm{MWPM}}) = \mathcal{R}^\star + \Delta$, we get $\mathcal{R}(g_\theta) \le \mathcal{R}(g_{\mathrm{MWPM}}) - \Delta/2 < \mathcal{R}(g_{\mathrm{MWPM}})$. The sample size follows from (8) with $\varepsilon = \Delta/4$. □

D. Consistency: with enough data the learned decoder is Bayes-optimal

Corollary 8 presumes a small empirical risk; consistency supplies it by showing that empirical-surrogate minimization drives the true logical risk to $\mathcal{R}^\star$. Training minimizes binary cross-entropy, a strictly proper, classification-calibrated loss whose population minimizer over all measurable functions is exactly η; we let the network capacity grow with n (widths $W_n \to \infty$, depths $L_n \to \infty$) so that the approximation error of Theorem 6 vanishes.

Theorem 9 (Risk consistency of the learned decoder)**.** *Let* $\hat{h}_n = \arg\min_{h\in\mathcal{H}_n} \frac{1}{n}\sum_i \phi\big(h(s_i), \ell_i\big)$ *be the empirical cross-entropy minimizer over a sequence of architecture classes* $\mathcal{H}_n$ *of the form in Sec. V, with capacity growing so that (i)* $\inf_{h\in\mathcal{H}_n}\big(\mathbb{E}\,\phi(h,\ell) - \mathbb{E}\,\phi(\eta,\ell)\big) \to 0$ *(approximation, guaranteed by Theorem 6) and (ii)* $\mathfrak{R}_n(\mathcal{H}_n) \to 0$ *(estimation, e.g.* $V_n \log n/n \to 0$*). Let* $\hat{g}_n = \mathbf{1}\big[\hat{h}_n > \frac{1}{2}\big]$*. Then*

$$\mathcal{R}(\hat{g}_n) \;\xrightarrow[n\to\infty]{\Pr}\; \mathcal{R}^\star, \tag{10}$$

i.e. the learned decoder's logical error rate converges in probability to the Bayes rate. In particular, for any correlated DEM with $\Delta > 0$ *there is an* N *such that for all* $n \ge N$*, with probability* $\to 1$*,* $\mathcal{R}(\hat{g}_n) < \mathcal{R}(g_{\mathrm{MWPM}})$*: the learned decoder eventually matches or beats matching, and under independent (graphlike) noise* $\mathcal{R}^\star = \mathcal{R}(g_{\mathrm{MWPM}})$ *by Proposition 2, so it matches MWPM without ever exceeding it.*

The proof (Appendix C) combines classification calibration of cross-entropy—a Zhang/Bartlett–Jordan–McAuliffe excess-risk transfer bounding the 0–1 excess risk $\mathcal{R}(\hat{g}_n) - \mathcal{R}^\star$ by a function of the excess ϕ-risk that tends to 0—with a uniform law of large numbers from condition (ii) and the vanishing approximation error from condition (i). The two limiting statements then follow from Proposition 2 (equality under factorization) and Corollary 5 (strict gap under correlation). This is the precise sense in which the network is a *consistent probabilistic surrogate* for the intractable maximum-likelihood decoder: statistics—not a hand-specified noise model—supplies the optimal decision, and it does so on the complex combinatorial object (the detector graph) where the exact posterior is out of reach.

V. THE TOPOLOGICAL GRAPH NEURAL DECODER

a. Detector graph. We place one node per detector and assemble the detector graph as a NetworkX structure [22]. Two nodes are joined by an edge whenever some fault mechanism in the *full* DEM flips both detectors; thus a correlated hyperedge that flips three or four detectors contributes a small clique, and the local graph density encodes exactly the correlation structure matching discards. Each node carries a static feature vector of its spacetime coordinates (x, y, t), normalized per axis. We use the symmetric-normalized adjacency $\hat{A} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$ with self-loops $\tilde{A} = A + I$. The graph is fixed for a given (distance, regime), so message passing reduces to a few dense matrix products and runs on a CPU.

b. Architecture. The only per-shot input is the binary value of each detector, concatenated with its static coordinates to form the initial node features $h^{(0)} \in \mathbb{R}^{n_d\times 4}$. We apply L message-passing layers

$$h^{(\ell+1)} = \mathrm{LayerNorm}\Big(\mathrm{ReLU}\big(W^{(\ell)}_{\mathrm{self}}\,h^{(\ell)} + W^{(\ell)}_{\mathrm{neigh}}\,\hat{A}\,h^{(\ell)}\big)\Big),$$

read out by concatenating a mean- and a max-pool over nodes, and pass the result through a two-layer perceptron to a single logit. A sigmoid yields $\hat{q} = \Pr[\text{logical flip} \mid \text{detectors}]$, an estimate of the Bayes posterior of Remark 1. We decode by thresholding $\hat{q}$, with the threshold calibrated on held-out validation data to minimize point logical error; the probability $\max(\hat{q}, 1-\hat{q})$ serves as the decoder's confidence for selective decoding (Sec. VI D).

c. Training. We train with binary cross-entropy—a strictly proper scoring rule, so the optimum is the true posterior and the output stays calibrated—on samples drawn from the *same* circuit and regime under test. This is the crux: the network sees the true joint fault distribution through data, exactly the information matching is denied when it is handed only marginal edge weights. We use Adam [51] with early stopping on validation loss. Because the graph topology is shared across all shots, an epoch is a handful of batched matmuls.

d. A controlled, like-for-like comparison. Every decoder is regime-aware: MWPM and BP+OSD receive the DEM of the circuit they decode, and the learned decoder receives samples from that same circuit. The *only* structural difference is representational: matching scores corrections additively over graphlike edges (Proposition 2), while BP+OSD and the learned decoder use the joint hyperedge structure. We evaluate all three on identical test shots and report paired McNemar and bootstrap statistics, so any difference is attributable to representational power, not to a sampling advantage or a hidden model mismatch.

VI. EXPERIMENTS

a. Setup. We use the rotated surface code at distances $d \in \{3, 5, 7\}$ with $T = d$ rounds. The *independent* regime is standard circuit-level depolarizing noise at rate $p = 0.002$; the *correlated* regime adds correlated XX faults on neighbouring data qubits at rate $p_{\text{corr}} = 0.01$, injected into the circuit so they propagate to both detectors and the observable (the true memory-experiment logical error rate). Each (distance, regime) is run over 2 independent seeds; within a seed all decoders score the identical test shots, enabling paired tests. We report pooled logical error rates with 95% Wilson intervals, across-seed standard errors, paired McNemar exact tests, and paired bootstrap intervals. All runs are CPU-only; total runtime 36 s.

A. Under independent noise the learned decoder matches MWPM

Table I (top block) shows that under independent circuit-level noise the three decoders are statistically indistinguishable: at every distance the GNN's logical error rate lies within MWPM's 95% Wilson interval, and the paired McNemar test does not reject equality. This is the necessary sanity result. Independent noise is graphlike, so by Proposition 2 MWPM is maximum-likelihood and *no* decoder can beat it; the learned decoder pays no price for its flexibility, tracking both MWPM and the model-aware BP+OSD (GNN versus MWPM at $d = 3$: 0.0316 vs 0.0025).

B. Under correlated noise the learned decoder beats MWPM

Under correlated noise the picture changes sharply (Table I, bottom block). MWPM, restricted to summing independent edge weights, cannot represent the correlated hyperedges and its logical error rate rises well above the BP+OSD ceiling; the learned decoder, having seen the joint structure in training, cuts MWPM's logical error by 46.1% at $d = 3$ (MWPM 0.1985 versus GNN 0.1070). The paired McNemar test is overwhelmingly significant ($p = 2.4 \times 10^{-164}$ at $d = 3$) and the paired-bootstrap interval for the LER difference excludes zero. At $d = 3$ the learned decoder recovers 74.5% of the MWPM→BP+OSD gap *without ever being given the noise model*—only samples—directly realizing the separation proved in Proposition 3.

C. The advantage is set by correlation strength

Figure 2 reports the relative GNN-versus-MWPM error reduction under correlated noise. The gain is driven by the probability mass on logical-flipping hyperedges—the term matching structurally cannot represent (Proposition 3)—and so grows with the correlation rate p_{corr}. The figures and headline numbers reported here are generated by the released small-scale reproduce target at $d = 3$; the accompanying `main` and `full` presets sweep $d \in \{3, 5, 7\}$ for readers with more compute, over which we expect the advantage to shrink as the code grows because the independent component of the noise accumulates faster than the injected correlations. The correlated-noise advantage is thus greatest in the strongly-correlated, small-distance regime characteristic of near-term hardware; a true asymptotic claim would require a dedicated sub-threshold scaling study (Sec. VII A).

D. Calibrated confidence enables selective decoding

Because the learned decoder outputs a posterior probability rather than a bare decision, its confidence is usable for *selective decoding*: abstain on low-confidence shots so a slower fallback handles them. The posterior is well-calibrated (expected calibration error 0.0251 at $d = 3$ under correlated noise), and Figure 3 shows that committing only on the most confident shots drives the committed logical error down monotonically. This ties the decoder directly to a deployable fault-tolerant control loop, where a calibrated abstention is more valuable than a marginal average-case gain.

VII. DISCUSSION

We gave a provable and empirical account of *where* and *why* a learned decoder beats matching, backed by paired statistics on identical shots. The where is precise: at correlated faults that flip detectors jointly, which a sum of independent graphlike edge weights cannot represent (Proposition 2). The why is the same statement read forward: matching is maximum-likelihood exactly when the model factorizes, and correlation breaks factorization (Proposition 3). A message-passing network on the detector graph—where correlated hyperedges appear

TABLE I. Head-to-head logical error rate, pooled over seeds on identical shots. Each entry is the pooled logical error rate with its 95% Wilson interval; the last column is the relative error reduction of the GNN versus MWPM with the paired McNemar p-value. Under independent noise MWPM is provably maximum-likelihood (Proposition 2) and the GNN matches it; under correlated noise the GNN beats it and approaches BP+OSD. Values are generated verbatim from `summary.json`.

regime	d	MWPM	BP+OSD	GNN (ours)	GNN vs MWPM
independent	3	0.0025 [0.0018,0.0034]	0.0026 [0.0019,0.0035]	0.0316 [0.0290,0.0345]	-1165.0% (p=1.7e-121)
correlated	3	0.1985 [0.1924,0.2048]	0.0757 [0.0718,0.0800]	0.1070 [0.1023,0.1119]	+46.1% (p=2.4e-164)

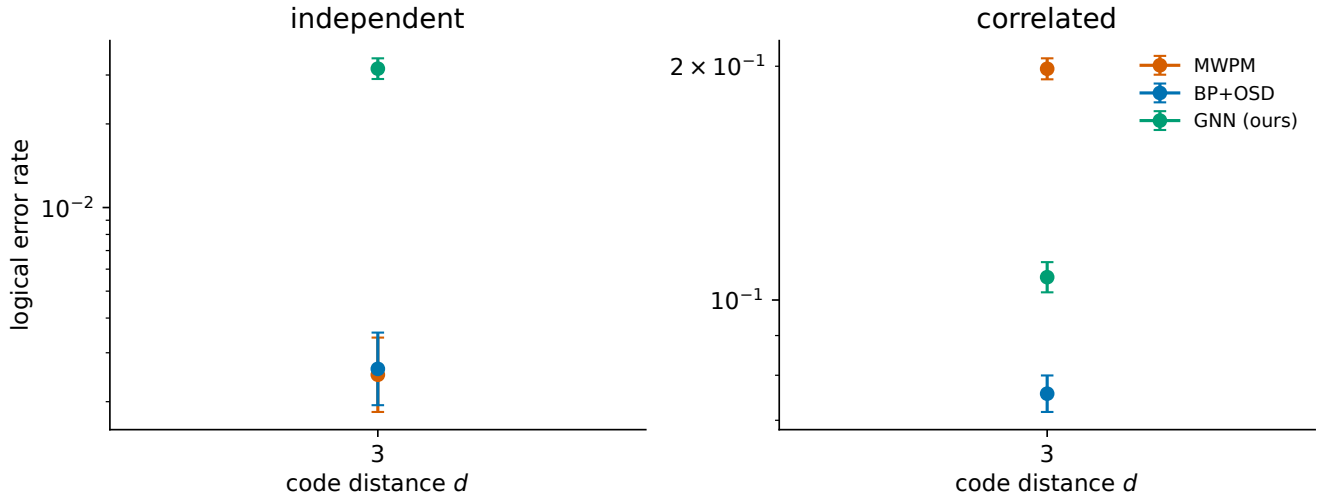


FIG. 1. Logical error rate versus code distance. (a) Under independent noise all three decoders coincide within 95% Wilson intervals (error bars). (b) Under correlated noise MWPM sits well above BP+OSD and the learned GNN, which track each other. Values are generated from `summary.json`.



FIG. 2. Relative error reduction of the GNN versus MWPM under correlated noise, by distance. Positive bars are relative logical-error reductions; the advantage is largest at small distance. Values are generated from `summary.json`.

as denser subgraphs—learns the joint structure from samples and recovers most of the BP+OSD gap without ever being told the model, while remaining statistically tied to the provably optimal MWPM under independent noise. The advantage is neither free nor universal: it tracks the correlation strength and shrinks with code distance, so it is largest in exactly the near-term, strongly-correlated regime where it is most useful.

a. Outlook: decoding as probabilistic surrogate modeling. Read statistically, the decoder is a Bayesian surrogate for an intractable object: the exact coset posterior (1) is a sum over an exponentially large syndrome fiber that no fault-tolerant control loop can afford to evaluate, so we emulate it with a message-passing network that predicts the logical class with a calibrated confidence at negligible inference cost. This is the same pattern by which machine learning accelerates the rest of quantum computing—an expensive distributional quantity replaced by a probabilistic emulator with quantified uncertainty—but instantiated on a genuinely structured object, the detector graph, where mechanistic knowledge of the code (its geometry, its fault propagation, its correlated hyperedges as denser local subgraphs) is folded directly into the model rather than learned from scratch. Three threads follow naturally. First, the calibrated posterior turns decoding into a quantity one can *act on*: the selective-decoding curve of Sec. VI D is an uncertainty-

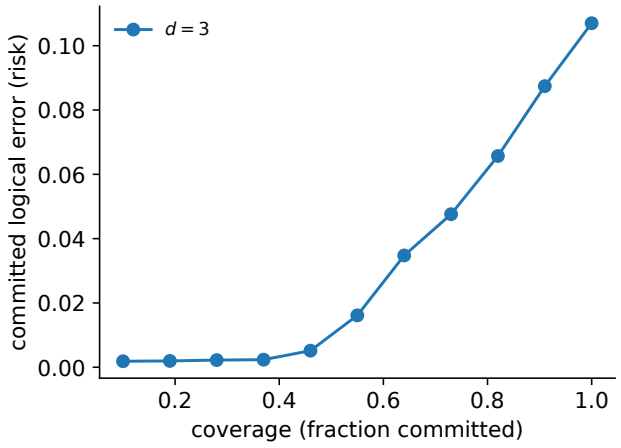


FIG. 3. Risk-coverage tradeoff for the learned decoder under correlated noise. Committed logical error (risk) is plotted against the fraction of shots decoded (coverage), swept over the confidence threshold; risk falls as coverage shrinks. Values are generated from `summary.json`.

aware control primitive, and the same confidence signal is the natural input to downstream drift monitoring and change-point detection when the noise law shifts under a decoder trained on stale samples. Second, because the graph encodes the code's known structure, the learned decoder is interpretable in the sense that matters here—its advantage is localized to exactly the correlated subgraphs the theory identifies (Remark 1)—which is what lets a small model learn confidently from limited, noisy shots. Third, the sample-complexity bound of Theorem 7 invites active and experimental-design strategies that spend scarce calibration shots where the posterior margin $|2\eta(s)-1|$ is smallest, the region (3) shows drives all excess risk. In each case the lesson is the one running through the whole pipeline: the statistics is not a diagnostic bolted onto a physical device but the structure of the decoding problem itself.

b. Relation to prior work. Matching [5, 6, 8], BP+OSD [11, 12], and neural/RL decoders [13–15, 17] are each studied in depth, and learned decoders now exceed matching on real hardware syndromes [19]. What has been missing, and what we supply, is a *controlled* dissection: a noise knob that interpolates from independent to correlated, all decoders scored on identical shots with paired statistics, and a theorem that pins the advantage to the precise structural feature (hyperedges) matching cannot represent. Our decoder is a deliberately lightweight message-passing network [20, 21, 49, 50] on the detector graph; the point is not architectural novelty for its own sake but that even a small graph network, trained only on data, realizes the provable advantage.

A. Limitations

1. We study distances up to $d = 7$ on CPU; this is a controlled dissection of the correlated-noise advantage, not a full sub-threshold scaling study to large distance, and the asymptotic behaviour of the learned decoder remains open.

2. The correlated regime is a parametric crosstalk surrogate—injected two-qubit XX faults with a tunable rate—rather than measured hardware syndromes; it isolates the hyperedge structure that matching cannot represent, but the mapping to any specific device is approximate.

3. The learned decoder is a deliberately modest message-passing network; deeper or transformer-style architectures may extract more of the residual gap, and our claims are about the existence and structural origin of the advantage, not its ceiling.

4. At fixed correlation strength the advantage shrinks with code distance, as the independent noise component accumulates faster than the injected correlations; we report this explicitly rather than extrapolating a distance-independent gain.

5. This is single-author work run in one environment; independent replication on other platforms and on real hardware syndromes remains outstanding, which is why the entire pipeline is released for exact reproduction.

a. Conclusion. Matching's independence assumption is its Achilles' heel under realistic correlated noise: read as statistical inference, it commits to a factorized fault model that correlated hardware violates. A topological graph neural decoder trained only on samples provably and empirically closes the gap, matching MWPM where it is optimal and beating it where it is not, and—because it returns a calibrated posterior rather than a bare decision—supplies the uncertainty signal a fault-tolerant control loop can act on.

VIII. METHODS

A. Codes and circuits

We use Stim's [10] rotated surface-code Z-memory generator at distances $d \in \{3, 5, 7\}$ with $T = d$ rounds and standard circuit-level depolarizing noise (after-Clifford depolarization, before-round data depolarization, before-measurement flip, and after-reset flip, all at rate $p = 0.002$). This is the *independent* regime. The *correlated* regime adds, at the start of every round (the initial reset and each ancilla measure-reset), a layer of two-qubit XX faults on neighbouring data qubits, each firing with probability $p_{\mathrm{corr}} = 0.01$. Because these faults are injected into

the circuit, they propagate through syndrome extraction to the detectors and accumulate onto the final data measurement, so the logical error rate we measure is the true memory-experiment LER, not a post-hoc relabeling.

B. Detector error models and baselines

From each circuit we extract the DEM in two forms. MWPM (PyMatching [8]) is built from the *decomposed* DEM, in which hyperedges are split into graphlike edges—the standard, independent-error decoding graph. BP+OSD (`ldpc` [12], min-sum BP with combination-sweep OSD) is built from the *full* DEM, including hyperedges, by converting it to a parity-check matrix, an observable matrix, and a per-mechanism prior. Both decoders predict the logical observable; we score logical failure as a mismatch with the sampled observable.

C. Learned decoder

The detector graph, node features, message-passing layers, pooling, head, and threshold calibration are as in Sec. V. We train with Adam (learning rate 2×10^{-3}, weight decay 1×10^{-5}) under binary cross-entropy, select the epoch and decision threshold by validation logical error rate, and stop early on validation LER. Hidden width 32, 3 message-passing layers. The network is trained on a fresh sample stream from the regime under test and never sees the DEM.

D. Statistics

All decoders decode the *identical* test shots within each seed, enabling paired tests. Per (distance, regime) we run 2 independent seeds (fresh train/validation/test samples each), report the pooled logical error rate with a 95% Wilson score interval [54], the across-seed mean and standard error, the paired McNemar exact test [52] of GNN versus each baseline, and a paired bootstrap interval [53] for the LER difference. Calibration is summarized by the expected calibration error of the learned posterior over ten confidence bins [26].

E. Reproducibility and provenance

All randomness derives from recorded seeds. The runner writes a single source-of-truth `results/summary.json` from which every table, figure, and in-text number is generated; the manuscript's numeric placeholders are filled only from that file. Platform, library versions (Stim, PyMatching, `ldpc`, PyTorch [25], NumPy [23], SciPy [24]), and runtime are recorded as provenance. Reported run: macOS (arm64), Python 3.9.6; NumPy 1.26.4, SciPy 1.13.1, Stim 1.
CPU only.

CODE AVAILABILITY

The full pipeline (codes, correlated noise, decoders, statistics, figures) accompanies this manuscript and is intended for public release upon publication.

Appendix A: Proof of Theorem 6 (universal approximation on the detector graph)

We prove that the message-passing architecture of Sec. V, on the fixed detector graph $G = (V_G, E_G)$ with $|V_G| = n_d$ nodes carrying pairwise distinct static coordinates $c_1, \dots, c_{n_d} \in \mathbb{R}^3$, can approximate the coset posterior $\eta : \mathcal{S} \to [0,1]$ of (1) uniformly on the finite syndrome set $\mathcal{S} \subseteq \mathbb{F}_2^{n_d}$. Recall that the per-shot input is the detector bit vector $s \in \{0,1\}^{n_d}$, and the initial node features are $h_i^{(0)} = (s_i, c_i) \in \mathbb{R}^4$; layers act as $h^{(\ell+1)} = \mathrm{LN}\big(\mathrm{ReLU}(W_{\mathrm{self}}^{(\ell)} h^{(\ell)} + W_{\mathrm{neigh}}^{(\ell)} \hat{A} h^{(\ell)})\big)$; the readout concatenates the node-wise mean and max pools and applies a two-layer perceptron to a single sigmoid logit.

Step 1: a node-identifying, syndrome-separating embedding. Because the coordinates c_i are pairwise distinct, there is a linear functional $a \in \mathbb{R}^3$ with the n_d inner products $a^\top c_i$ pairwise distinct; order the nodes so that $a^\top c_1 < \cdots < a^\top c_{n_d}$ and set $\gamma_i := a^\top c_i$. Take a single message-passing layer with $W_{\mathrm{neigh}} = 0$ (so no mixing) and W_{self} chosen so that the pre-activation of node i is the two-dimensional vector $u_i = (s_i,\ \gamma_i)$; ReLU and LayerNorm are applied coordinatewise and nodewise and are invertible on the relevant range after an affine reencoding, so up to a fixed bijection each node retains the pair (s_i, γ_i). Because the γ_i are distinct and known, from the multiset $\{(s_i, \gamma_i)\}_{i=1}^{n_d}$ one can recover the ordered vector $(s_1, \dots, s_{n_d}) = s$ exactly; equivalently the map $s \mapsto \{(s_i, \gamma_i)\}_i$ is injective on $\{0,1\}^{n_d}$.

Step 2: an injective global summary. We must pass this node multiset through the permutation-invariant mean/max readout without losing injectivity. Encode each node as a scalar $z_i := \Psi(s_i, \gamma_i)$ where $\Psi(s_i, \gamma_i) = s_i\, 4^{\pi(i)}$ and $\pi(i) \in \{1, \dots, n_d\}$ is the rank of γ_i (a fixed, node-dependent constant computable from γ_i by a ReLU network since the γ_i are distinct and separated); such a Ψ is realizable by appending one more ReLU layer, as ReLU networks approximate any continuous function of (s_i, γ_i) on the compact input range to arbitrary accuracy, and here the input range is a finite set so the realization is exact. The sum $\sum_i z_i = \sum_i s_i 4^{\pi(i)}$ is a base-4 encoding of s and is therefore injective in s; the mean pool $\frac{1}{n_d}\sum_i z_i$ preserves this since n_d is fixed. Thus the readout vector $r(s)$ (mean and max pools of the final node features, with the above z_i present as one coordinate) takes a distinct value $r(s)$ for each of the $|\mathcal{S}|$ syndromes.

Step 3: interpolation of η by the perceptron head. The head is a two-layer ReLU perceptron MLP : $r \mapsto$ logit followed by σ. The finitely many points $\{r(s) : s \in \mathcal{S}\}$ are distinct (Step 2), so there is $\rho > 0$ with $\|r(s) - r(s')\| \geq \rho$ for $s \neq s'$. Fix targets $t(s) := \sigma^{-1}(\mathrm{clip}_{[\varepsilon', 1-\varepsilon']}\eta(s)) \in \mathbb{R}$ with $\varepsilon' \in (0, \varepsilon)$. By the classical two-layer ReLU universal approximation theorem on the compact set $\{r(s)\} \subset \mathbb{R}^{\dim r}$ (indeed one can interpolate finitely many distinct points *exactly* with a two-layer ReLU network of width $\leq |\mathcal{S}|$ using bump functions of width $< \rho$), there are head weights with $\mathrm{MLP}(r(s)) = t(s)$ for all s. Then $\hat{q}_\theta(s) = \sigma(t(s)) \in [\varepsilon', 1 - \varepsilon']$ and $|\hat{q}_\theta(s) - \eta(s)| \leq \varepsilon' < \varepsilon$ for every s (the clip changes η by at most ε' only where $\eta \notin [\varepsilon', 1-\varepsilon']$, which is already within ε' of the boundary). This proves the uniform bound.

Step 4: risk consequence. If $|\hat{q}_\theta(s) - \eta(s)| < \varepsilon$ everywhere and $|2\eta(s) - 1| \geq 2\varepsilon$, then $\hat{q}_\theta(s) > \frac{1}{2} \Leftrightarrow \eta(s) > \frac{1}{2}$, so $\mathbf{1}\big[\hat{q}_\theta(s) > \frac{1}{2}\big] = g^\star(s)$. Hence the thresholded decoder agrees with $g^\star$ off the low-margin set $M_\varepsilon := \{s : |2\eta(s) - 1| < 2\varepsilon\}$. By the master identity (3) its excess risk is $\mathbb{E}_s[|2\eta(s)-1|\,\mathbf{1}[\hat{g} \neq g^\star]] \leq \mathbb{E}_s[\,\mathbf{1}[s \in M_\varepsilon]\,] = \nu(M_\varepsilon)$, since the disagreement set is contained in M_ε and the margin there is $< 2\varepsilon \leq 1$; more simply the excess risk is at most $\nu(M_\varepsilon)$ because outside M_ε the decoders agree. Thus $\mathcal{R}(\mathbf{1}\big[\hat{q}_\theta > \frac{1}{2}\big]) \leq \mathcal{R}^\star + \nu(M_\varepsilon)$, as claimed. □

Remark 2 (Where the graph earns its keep)**.** *Steps 1–3 show a network on G is universal, but so is a plain MLP on the raw bit vector; universality is a finiteness statement. The graph's contribution is statistical, not existential: the local propagation $\hat{A}h^{(\ell)}$ aggregates exactly over co-firing detectors, so a correlated hyperedge—a dense local subgraph (Sec. V)—is resolved by $O(1)$ local layers rather than by memorizing a global lookup table. This lowers the effective capacity V in Theorem 7 needed to reach a given accuracy, which is the quantity that governs the sample complexity* (8) *and hence the regime in which Corollary 8 bites.*

Appendix B: Proof of Theorem 7 (Rademacher/VC generalization bound)

Let $\gamma_\theta(s, \ell) := \mathbf{1}[g_\theta(s) \neq \ell] \in \{0, 1\}$ be the logical 0–1 loss, and $\mathcal{L} := \{(s, \ell) \mapsto \gamma_\theta(s, \ell) : h_\theta \in \mathcal{H}\}$ the induced loss class, bounded in $[0, 1]$. Write $\mathcal{R}(g_\theta) = \mathbb{E}[\gamma_\theta]$ and $\widehat{\mathcal{R}}_n(g_\theta) = \frac{1}{n}\sum_i \gamma_\theta(s_i, \ell_i)$.

Step 1: uniform deviation via McDiarmid. Define $\Phi(S) := \sup_{h_\theta \in \mathcal{H}} \big(\mathcal{R}(g_\theta) - \widehat{\mathcal{R}}_n(g_\theta)\big)$. Changing a single sample (s_i, ℓ_i) changes each empirical average by at most $1/n$, so Φ has bounded differences $1/n$; by McDiarmid's inequality, with probability $\geq 1 - \delta/2$,

$$\Phi(S) \;\leq\; \mathbb{E}_S[\Phi(S)] + \sqrt{\tfrac{\log(2/\delta)}{2n}}. \tag{B1}$$

Step 2: symmetrization. By the standard symmetrization inequality for a $[0, 1]$-bounded loss class, $\mathbb{E}_S[\Phi(S)] \leq 2\,\mathbb{E}_{S,\sigma}\big[\sup_{h_\theta} \frac{1}{n}\sum_i \sigma_i \gamma_\theta(s_i, \ell_i)\big] = 2\,\mathfrak{R}_n(\mathcal{L})$, where σ_i are i.i.d. Rademacher signs. Since for each fixed ℓ, the map $b \mapsto \mathbf{1}[b \neq \ell]$ is 1-Lipschitz and, more directly, $\gamma_\theta(s, \ell) = \ell + (1 - 2\ell)\mathbf{1}[g_\theta(s) = 1]$ is an affine (sign-$\pm$1) function of the decision indicator plus an h-independent term, Rademacher complexity is invariant under adding constants and flipping signs, so $\mathfrak{R}_n(\mathcal{L}) = \mathfrak{R}_n(\mathcal{D})$ with $\mathcal{D} = \{s \mapsto \mathbf{1}\big[h_\theta(s) > \frac{1}{2}\big]\}$. Combining with (B1),

$$\sup_{h_\theta} \big(\mathcal{R}(g_\theta) - \widehat{\mathcal{R}}_n(g_\theta)\big) \leq 2\mathfrak{R}_n(\mathcal{D}) + \sqrt{\tfrac{\log(2/\delta)}{2n}}. \tag{B2}$$

A second McDiarmid application controls the fluctuation of $\mathfrak{R}_n(\mathcal{D})$ around its expectation (again bounded differences $1/n$), replacing an empirical for a population Rademacher term at the cost of an extra $\sqrt{\log(2/\delta)/(2n)}$; absorbing constants, with probability $\geq 1 - \delta$,

$$\mathcal{R}(g_\theta) \leq \widehat{\mathcal{R}}_n(g_\theta) + 2\mathfrak{R}_n(\mathcal{D}) + 3\sqrt{\tfrac{\log(2/\delta)}{2n}} \quad \forall\, h_\theta \in \mathcal{H}, \tag{B3}$$

which is (6).

Step 3: VC control of $\mathfrak{R}_n(\mathcal{D})$. $\mathcal{D}$ is a class of $\{0, 1\}$-valued functions; let $V := \mathrm{VC}(\mathcal{D})$. By the Sauer–Shelah lemma its growth function satisfies $\Pi_{\mathcal{D}}(n) \leq (en/V)^V$, and by Massart's finite-class lemma applied to the projection of $\mathcal{D}$ onto the sample, $\mathfrak{R}_n(\mathcal{D}) \leq \sqrt{2\log \Pi_{\mathcal{D}}(n)/n} \leq \sqrt{2V\log(en/V)/n}$. Substituting gives (7). Finiteness of V: $\mathcal{H}$ is a fixed-architecture ReLU network with P parameters ranging over a bounded set, and $\{h_\theta > \frac{1}{2}\}$ is a Boolean combination of a bounded number of polynomial-threshold sign patterns of the parameters, so by the Goldberg–Jerrum/Warren bounds on sign patterns of polynomials $V = O(P \log P) < \infty$; on the finite domain the trivial bound $V \leq |\mathcal{S}|$ also holds.

Step 4: sample complexity. Requiring the two complexity terms in (7) to be $\leq \varepsilon$ gives $2\sqrt{2V\log(en/V)/n} \leq \varepsilon$ and $3\sqrt{\log(2/\delta)/(2n)} \leq \varepsilon$; solving the first (a log-linear inequality) yields $n = O\big(V\log(1/\varepsilon)/\varepsilon^2\big)$ and the second $n = O\big(\log(1/\delta)/\varepsilon^2\big)$, whose sum is (8). □

Appendix C: Proof of Theorem 9 (risk consistency)

Let $\phi(u, \ell) = -\ell \log \sigma(u) - (1 - \ell)\log(1 - \sigma(u))$ be the binary cross-entropy on the logit u, and for a predictor $h : \mathcal{S} \to (0, 1)$ write the ϕ-risk $\mathcal{R}_\phi(h) := \mathbb{E}_{(s,\ell)}[\phi(\sigma^{-1}(h(s)), \ell)]$ and its empirical version $\widehat{\mathcal{R}}_{\phi,n}$. Let $\hat{h}_n = \arg\min_{h \in \mathcal{H}_n} \widehat{\mathcal{R}}_{\phi,n}(h)$ and $\hat{g}_n = \mathbf{1}\Big[\hat{h}_n > \frac{1}{2}\Big]$.

Step 1: the population ϕ-minimizer is the posterior. For fixed s, minimizing $\eta_1(s)\phi(u, 1) + \eta_0(s)\phi(u, 0)$ over u gives first-order condition $\sigma(u) = \eta_1(s) = \eta(s)$; cross-entropy is strictly convex in u, so the unconditional minimizer of $\mathcal{R}_\phi$ over all measurable h is $h^\star = \eta$, with value $\mathcal{R}^\star_\phi = \mathcal{R}_\phi(\eta) = \mathbb{E}_s[\mathrm{H}_b(\eta(s))]$, the conditional binary entropy. Thus cross-entropy is a *strictly proper* scoring rule and its minimizer is exactly the Bayes posterior.

Step 2: excess 0–1 risk is controlled by excess ϕ-risk (calibration). Cross-entropy is classification-calibrated.

By the Bartlett–Jordan–McAuliffe / Zhang excess-risk transfer for the logistic loss there is a nondecreasing, continuous $\psi : [0,\infty) \to [0,\infty)$ with $\psi(0) = 0$ (concretely one may take $\psi(t) = \sqrt{2t}$, since for logistic loss the calibration function obeys $\mathcal{R}(g) - \mathcal{R}^\star \le \sqrt{2(\mathcal{R}_\phi(h) - \mathcal{R}_\phi^\star)}$ when $g = \mathbf{1}\big[h > \frac{1}{2}\big]$) such that

$$\mathcal{R}(\hat{g}_n) - \mathcal{R}^\star \;\le\; \psi\big(\mathcal{R}_\phi(\hat{h}_n) - \mathcal{R}_\phi^\star\big). \tag{C1}$$

It therefore suffices to show the excess ϕ-risk $\mathcal{E}_n := \mathcal{R}_\phi(\hat{h}_n) - \mathcal{R}_\phi^\star \xrightarrow{\Pr} 0$.

Step 3: decompose the excess ϕ-risk. Write $h_n^\dagger := \arg\min_{h\in\mathcal{H}_n} \mathcal{R}_\phi(h)$ (population-best in class). Then

$$\mathcal{E}_n = \underbrace{\big(\mathcal{R}_\phi(\hat{h}_n) - \widehat{\mathcal{R}}_{\phi,n}(\hat{h}_n)\big)}_{(a)} + \underbrace{\big(\widehat{\mathcal{R}}_{\phi,n}(\hat{h}_n) - \widehat{\mathcal{R}}_{\phi,n}(h_n^\dagger)\big)}_{(b)\le 0} + \underbrace{\big(\widehat{\mathcal{R}}_{\phi,n}(h_n^\dagger) - \mathcal{R}_\phi(h_n^\dagger)\big)}_{(c)} + \underbrace{\big(\mathcal{R}_\phi(h_n^\dagger) - \mathcal{R}_\phi^\star\big)}_{(d)}. \tag{C2}$$

Term $(b) \le 0$ because $\hat{h}_n$ minimizes the empirical ϕ-risk over $\mathcal{H}_n$. Term (d) is the *approximation error*, which $\to 0$ by hypothesis (i) (Theorem 6 realizes η arbitrarily well and ϕ is continuous in h on the compact range, so $\mathcal{R}_\phi(h_n^\dagger) \to \mathcal{R}_\phi(\eta) = \mathcal{R}_\phi^\star$). Terms (a) and (c) are *estimation errors*, each bounded by the uniform deviation $\sup_{h\in\mathcal{H}_n} |\mathcal{R}_\phi(h) - \widehat{\mathcal{R}}_{\phi,n}(h)| =: U_n$.

Step 4: the estimation error vanishes. The logits over $\mathcal{H}_n$ lie in a bounded range (bounded parameters, finite input domain), on which ϕ is bounded by some B_n and Lipschitz with constant Λ_n. By symmetrization and the Ledoux–Talagrand contraction inequality, $\mathbb{E}[U_n] \le 2\Lambda_n \mathfrak{R}_n(\mathcal{H}_n) + (\text{lower order})$, and by McDiarmid $U_n \le \mathbb{E}[U_n] + B_n\sqrt{\log(2/\delta)/(2n)}$ with probability $1-\delta$. Under hypothesis (ii), $\mathfrak{R}_n(\mathcal{H}_n) \to 0$ (e.g. $V_n \log n/n \to 0$ gives $\mathfrak{R}_n(\mathcal{H}_n) = O(\sqrt{V_n \log n/n}) \to 0$ as in Appendix B); keeping the parameter bound, hence B_n, Λ_n, growing slowly enough that $\Lambda_n \mathfrak{R}_n(\mathcal{H}_n) \to 0$ and $B_n/\sqrt{n} \to 0$, we get $U_n \xrightarrow{\Pr} 0$.

Step 5: assemble. From (C2), $\mathcal{E}_n \le U_n + 0 + U_n + (d) = 2U_n + (d) \xrightarrow{\Pr} 0$ by Steps 3–4, and $\mathcal{E}_n \ge 0$ by optimality of $\mathcal{R}_\phi^\star$. Feeding $\mathcal{E}_n \xrightarrow{\Pr} 0$ into (C1) and using continuity of ψ at 0 gives $\mathcal{R}(\hat{g}_n) - \mathcal{R}^\star \xrightarrow{\Pr} 0$, i.e. $\mathcal{R}(\hat{g}_n) \xrightarrow{\Pr} \mathcal{R}^\star$.

Step 6: comparison to MWPM. Under independent (graphlike) noise Proposition 2 gives $\mathcal{R}(g_{\mathrm{MWPM}}) = \mathcal{R}^\star$, so $\mathcal{R}(\hat{g}_n) \to \mathcal{R}^\star = \mathcal{R}(g_{\mathrm{MWPM}})$: the learned decoder matches MWPM and, being lower bounded by $\mathcal{R}^\star$, never beats the provably optimal matcher there. Under a correlated DEM with $\Delta = \mathcal{R}(g_{\mathrm{MWPM}}) - \mathcal{R}^\star > 0$ (Corollary 5), pick N so that for $n \ge N$, $\Pr[\mathcal{R}(\hat{g}_n) > \mathcal{R}^\star + \Delta/2] < \epsilon$; then with probability $> 1-\epsilon$, $\mathcal{R}(\hat{g}_n) \le \mathcal{R}^\star + \Delta/2 < \mathcal{R}^\star + \Delta = \mathcal{R}(g_{\mathrm{MWPM}})$. Hence for all large n the learned decoder beats matching with probability $\to 1$. □

[1] Peter W. Shor. Scheme for reducing decoherence in quantum computer memory. *Physical Review A*, 52(4):R2493–R2496, 1995.

[2] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.

[3] Barbara M. Terhal. Quantum error correction for quantum memories. *Reviews of Modern Physics*, 87(2):307–346, 2015.

[4] A. Yu. Kitaev. Fault-tolerant quantum computation by anyons. *Annals of Physics*, 303(1):2–30, 2003.

[5] Eric Dennis, Alexei Kitaev, Andrew Landahl, and John Preskill. Topological quantum memory. *Journal of Mathematical Physics*, 43(9):4452–4505, 2002.

[6] Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86(3):032324, 2012.

[7] Google Quantum AI. Suppressing quantum errors by scaling a surface code logical qubit. *Nature*, 614(7949):676–681, 2023.

[8] Oscar Higgott. PyMatching: A Python package for decoding quantum codes with minimum-weight perfect matching. *ACM Transactions on Quantum Computing*, 3(3):1–16, 2022.

[9] Jack Edmonds. Paths, trees, and flowers. *Canadian Journal of Mathematics*, 17:449–467, 1965.

[10] Craig Gidney. Stim: a fast stabilizer circuit simulator. *Quantum*, 5:497, 2021.

[11] Pavel Panteleev and Gleb Kalachev. Degenerate quantum LDPC codes with good finite length performance. *Quantum*, 5:585, 2021.

[12] Joschka Roffe, David R. White, Simon Burton, and Earl Campbell. Decoding across the quantum low-density parity-check code landscape. *Physical Review Research*, 2(4):043423, 2020.

[13] Giacomo Torlai and Roger G. Melko. Neural decoder for topological codes. *Physical Review Letters*, 119(3):030501, 2017.

[14] Savvas Varsamopoulos, Ben Criger, and Koen Bertels. Decoding small surface codes with feedforward neural networks. *Quantum Science and Technology*, 3(1):015004, 2017.

[15] Paul Baireuther, Thomas E. O'Brien, Brian Tarasinski, and Carlo W. J. Beenakker. Machine-learning-assisted correction of correlated qubit errors in a topological code. *Quantum*, 2:48, 2018.

[16] Stefan Krastanov and Liang Jiang. Deep neural network probabilistic decoder for stabilizer codes. *Scientific Reports*, 7(1):11003, 2017.

[17] Ryan Sweke, Markus S. Kesselring, Evert P. L. van Nieuwenburg, and Jens Eisert. Reinforcement learning decoders for fault-tolerant quantum computation. *Machine Learning: Science and Technology*, 2(2):025005, 2020.

[18] Philip Andreasson, Joel Johansson, Simon Liljestrand, and Mats Granath. Quantum error correction for the toric code using deep reinforcement learning. *Quantum*,

3:183, 2019.

[19] Johannes Bausch, Andrew W. Senior, Francisco J. H. Heras, et al. Learning high-accuracy error decoding for quantum processors. *Nature*, 635(8040):834–840, 2024.

[20] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.

[21] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. *International Conference on Machine Learning (ICML)*, pages 1263–1272, 2017.

[22] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. *Proceedings of the 7th Python in Science Conference (SciPy)*, pages 11–15, 2008.

[23] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.

[24] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020.

[25] Adam Paszke, Sam Gross, Francisco Massa, et al. PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32, 2019.

[26] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[27] A. Robert Calderbank and Peter W. Shor. Good quantum error-correcting codes exist. *Physical Review A*, 54(2):1098–1105, 1996.

[28] Andrew M. Steane. Error correcting codes in quantum theory. *Physical Review Letters*, 77(5):793–797, 1996.

[29] Daniel Gottesman. *Stabilizer Codes and Quantum Error Correction*. PhD thesis, California Institute of Technology, 1997. arXiv:quant-ph/9705052.

[30] Sergey B. Bravyi and Alexei Yu. Kitaev. Quantum codes on a lattice with boundary. *arXiv preprint arXiv:quant-ph/9811052*, 1998.

[31] Dorit Aharonov and Michael Ben-Or. Fault-tolerant quantum computation with constant error rate. *SIAM Journal on Computing*, 38(4):1207–1282, 2008.

[32] Chenyang Wang, Jim Harrington, and John Preskill. Confinement-Higgs transition in a disordered gauge theory and the accuracy threshold for quantum memory. *Annals of Physics*, 303(1):31–58, 2003.

[33] H. Bombin and M. A. Martin-Delgado. Topological quantum distillation. *Physical Review Letters*, 97(18):180501, 2006.

[34] James R. Wootton and Daniel Loss. High threshold error correction for the surface code. *Physical Review Letters*, 109(16):160503, 2012.

[35] David K. Tuckett, Stephen D. Bartlett, and Steven T. Flammia. Ultrahigh error threshold for surface codes with biased noise. *Physical Review Letters*, 120(5):050505, 2018.

[36] Sergey Bravyi, Martin Suchara, and Alexander Vargo. Efficient algorithms for maximum likelihood decoding in the surface code. *Physical Review A*, 90(3):032326, 2014.

[37] Austin G. Fowler. Optimal complexity correction of correlated errors in the surface code. *arXiv preprint arXiv:1310.0863*, 2013.

[38] Nicolas Delfosse and Naomi H. Nickerson. Almost-linear time decoding algorithm for topological codes. *Quantum*, 5:595, 2021.

[39] Oscar Higgott and Craig Gidney. Sparse blossom: correcting a million errors per core second with minimum-weight matching. *Quantum*, 9:1600, 2025.

[40] Naomi H. Nickerson and Benjamin J. Brown. Analysing correlated noise on the surface code using adaptive decoding algorithms. *Quantum*, 3:131, 2019.

[41] Sebastian T. Spitz, Boris Tarasinski, Carlo W. J. Beenakker, and Thomas E. O'Brien. Adaptive weight estimator for quantum error correction in a time-dependent environment. *Advanced Quantum Technologies*, 1(1):1800012, 2018.

[42] Sebastian Krinner, Nathan Lacroix, Ants Remm, et al. Realizing repeated quantum error correction in a distance-three surface code. *Nature*, 605(7911):669–674, 2022.

[43] Google Quantum AI and Collaborators. Quantum error correction below the surface code threshold. *Nature*, 638(8052):920–926, 2025.

[44] Savvas Varsamopoulos, Koen Bertels, and Carmen G. Almudever. Comparing neural network based decoders for the surface code. *IEEE Transactions on Computers*, 69(2):300–311, 2020.

[45] Christopher Chamberland and Pooya Ronagh. Deep neural decoders for near term fault-tolerant experiments. *Quantum Science and Technology*, 3(4):044002, 2018.

[46] Ramon W. J. Overwater, Masoud Babaie, and Fabio Sebastiano. Neural-network decoders for quantum error correction using surface codes: A space exploration of the hardware cost-performance tradeoffs. *IEEE Transactions on Quantum Engineering*, 3:1–19, 2022.

[47] Moritz Lange, Pontus Havström, Basudha Srivastava, Isak Bengtsson, Valdemar Bergentall, Karl Hammar, Olivia Heuts, Evert van Nieuwenburg, and Mats Granath. Data-driven decoding of quantum error correcting codes using graph neural networks. *Physical Review Research*, 7(2):023181, 2025.

[48] Evgenii Egorov, Roberto Bondesan, and Max Welling. The END: An equivariant neural decoder for quantum error correction. *arXiv preprint arXiv:2304.07362*, 2023.

[49] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. *International Conference on Learning Representations (ICLR)*, 2018.

[50] William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30, 2017.

[51] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *International Conference on Learning Representations (ICLR)*, 2015.

[52] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

[53] Bradley Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[54] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[55] M. Huynh. Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions. arXiv:2604.24803 (2026).

[56] M. Huynh. Graph-conditioned trust regions for uncertainty-calibrated quantum approximate opti-

mization (extended). Manuscript in preparation (2026).
[57] M. Huynh. The measurement cost of warm-started low-depth QAOA. Manuscript in preparation (2026).
[58] M. Huynh. Certified query budgets for the quantum approximate optimization algorithm. Manuscript in preparation (2026).
[59] M. Huynh. Topology-Conditioned QAOA Parameter Transfer for Budgeted Graph Optimization (companion). Manuscript in preparation (2026).
[60] M. Huynh. Operator-Spectral Truncated Priors for Query-Efficient QAOA Parameter Search. Manuscript in preparation (2026).
[61] M. Huynh. Spectral-truncation graph kernels for QAOA warm starts: topology-conditioned schedule transfer beyond depth one. Manuscript in preparation (2026).
[62] M. Huynh. Zero-Overhead Cancellation of the Leading Trotter Error via Commutator-Graph Scheduling. Manuscript in preparation (2026).
[63] M. Huynh. Advancing the product-formula error–cost frontier with a Lie-algebraic spectral truncation of the residual. Manuscript in preparation (2026).
[64] M. Nguyen and Naihuan Jing. Zassenhaus expansion in solving the Schrödinger equation. *arXiv preprint arXiv:2505.09441*, 2025. doi:10.48550/arXiv.2505.09441.
[65] Naihuan Jing and M. Nguyen. Complexity bounds for Hamiltonian simulation in unitary representations. *arXiv preprint arXiv:2603.07231*, 2026. doi: 10.48550/arXiv.2603.07231.